

Ethical Risks and Opportunities in Designing Artificial Awareness

Ana Tanevska, Sadegh Soudjani, Luis Fernando D'Haro, Jesús Salgado, Celia Fernández, Piera Maurizio, Ismael T. Freire, Raja Chatila, Michele Farisco, Kathinka Evers, Luis Marcos, Ginevra Castellano, Didem Gurdur Broo, Matthijs Smakman, Ophelia Deroy, Stephan Sigg, Nihan Kahraman, Michael Laakasuo, Manuel Gasulla, Xujun Ma, Martin Andraud, Giovanni Iacca, Serena Marchesi, Chris Register, Maryam Khan, Alberto Giubilini, Mehdi Khamassi

Ana Tanevska^[0000-0002-2628-4123]

Please complete Institute, Uppsala University, e-mail: ana.tanevska@it.uu.se

Sadegh Soudjani^[1111-2222-3333-4444]

Name, Address of Institute e-mail: sadegh@mpi-sws.org

Luis Fernando D'Haro

Name, Address of Institute e-mail: luisfernando.dharo@upm.es

Jesús Salgado

Name, Address of Institute e-mail: jesus.salgado@upm.es

Celia Fernández

Name, Address of Institute e-mail: mariacelia.fernandez@upm.es

Piera Maurizio

Name, Address of Institute e-mail: piera.maurizio@gmail.com

Michele Farisco, Kathinka Evers

Centre for Research Ethics and Bioethics, Uppsala University, Uppsala, Sweden e-mail: michele.farisco@uu.se, kathinka.evers@uu.se

Luis Marcos

Name, Address of Institute e-mail: lmarcos1@researchmar.net

Ginevra Castellano

Name, Address of Institute e-mail: ginevra.castellano@it.uu.se

Didem Gurdur Broo

Name, Address of Institute e-mail: didem.gurdur.broo@it.uu.se

Matthijs Smakman, Ophelia Deroy

Name, Address of Institute e-mail: matthijs.smakman@hu.nl, ophelia.deroy@lmu.de

Stephan Sigg

Name, Address of Institute e-mail: stephan.sigg@aalto.fi

Nihan Kahraman

Name, Address of Institute e-mail: nicoskun@yildiz.edu.tr

Michael Laakasuo

Name, Address of Institute e-mail: michael.laakasuo@helsinki.fi

Manuel Gasulla

Name, Address of Institute e-mail: manel.gasulla@upc.edu

Xujun Ma

Abstract With the rapid advancement of technology, artificial agents, systems, and robots are increasingly seen as interactive actors possessing some level of autonomy, as well as some scenario-dependent social abilities. Enhancing these systems additionally with the functionality of computational awareness brings them one step closer to autonomous artificial subjects, bringing with them a wide set of ethical and moral challenges. **If an artificial agent with some degree of awareness is perceived as an autonomous social being by the human with whom it is interacting, how are the various facets of its design and behavior impacting its human partner?** How can one foster a successful interaction with the people in the environment, maximizing their comfort, acceptance, and trust in the agent? **And on a broader level, how can the potential negative impacts of aware AI systems on human societies be minimized, while improving the opportunities to improve their ethical behavior?** It becomes obvious that designing and implementing artificial awareness in any technology comes with essential ethical and moral considerations. Are intentionality and goal-oriented behavior required for artificial agents and systems to align with human values? Do they have to be aware of the potential consequences of their actions or recommendations? Does this require or imply moral agency? How can existing guidelines on Trustworthy and Responsible AI be adapted to and integrated in research on artificial awareness? This chapter explores the various ethical risks and opportunities encountered by the different projects in the EU's Awareness Inside portfolio, the identified relevant actors, and the governance strategies to mitigate those risks and promote the opportunities while aligning with the existing Trustworthy AI and Ethical-by-design guidelines.

1 Introduction

As artificial systems continue to evolve, their capabilities increasingly mirror aspects of human behavior, such as interaction, social abilities, and to a certain extent autonomy. One of the most profound current progress in this area is the integration of

Name, Address of Institute e-mail: xujun.ma@telecom-sudparis.eu

Martin Andraud

Name, Address of Institute e-mail: martin.andraud@uclouvain.be

Giovanni Iacca

Name, Address of Institute e-mail: giovanni.iacca@unitn.it

Serena Marchesi

Name, Address of Institute e-mail: serena.marchesi@iit.it

Alberto Giubilini, Chris Register, Maryam Khan

Name, Address of Institute e-mail: alberto.giubilini@uehiro.ox.ac.uk

Mehdi Khamassi^[0000-0002-2515-1046], Ismael T. Freire^[0000-0002-5740-1513], Raja Chatila^[0000-0001-7822-0634]

Institute of Intelligent Systems and Robotics, Sorbonne University, Centre National de la Recherche Scientifique, F-75005 Paris, France e-mail: mehdi.khamassi@sorbonne-universite.fr

artificial awareness. The concept of artificial awareness bridges the gap between basic automation systems and systems that **can perceive and interpret their environment in a way that supports adaptive, goal-directed behaviors**. As artificial systems approach the threshold of perceived autonomy, they introduce complex ethical challenges that merit careful consideration. These challenges are particularly evident when systems are perceived as autonomous intentional social entities by their human counterparts.

The ethical questions surrounding artificial awareness are multi-faceted. How do we ensure that such systems act in ways that align with human values? What safeguards can be put in place to prevent systems from pursuing objectives that may compromise human well-being? Furthermore, the integration of artificial awareness brings us closer to the possibility of machines that not only make decisions but also engage in reasoning and reflection. This capacity to act autonomously and potentially with moral implications raises significant concerns about their role in society. Should these systems be endowed with moral agency? Would this increase their propensity to align with human values or not? **How can existing frameworks for Trustworthy AI be adapted to manage the ethical implications of artificial awareness?**

This chapter explores these ethical concerns and the broader implications of artificial awareness across various projects in the EU's Awareness Inside portfolio. It examines the potential impacts of artificial awareness on human society, the ethical risks and opportunities it presents, and the governance mechanisms and structures necessary to ensure the responsible development and deployment of these systems.

1.1 A collective attempt to define awareness in artificial systems

Awareness is a complex and multidimensional concept [8], with varying interpretations across different fields. In the context of artificial intelligence and robotics, it is often debated whether awareness refers to a state of consciousness, akin to human awareness and self-awareness, or whether it represents a more functional or process-based capability. There is a common agreement in the scientific community that the same way research on animal awareness helps clarify the notion of awareness (Birch), research on artificial awareness can also help: Because by construction we know what is in the machine, we can not only compare a robot's behavior to that of humans, but also precisely characterize the information processing mechanisms underlying the modeled cognitive functions. This can be summarized by the famous quote of physicist Richard Feynman: "What I cannot create, I do not understand".

Two broad approaches to defining awareness can be distinguished/have been identified throughout the different projects of the Awareness Inside portfolio, which we can tentatively call **strong** and **weak** definitions of awareness.

Strong Definition of Awareness: The strong definition of artificial awareness is typically associated with higher-level cognitive processes, often tied to theories of consciousness and higher-order awareness. The definition emphasizes the system's capacity to model and manipulate information internally in a way that supports advanced decision-making and goal-oriented behavior. In this sense, awareness is

not merely the tracking of stimuli or input-output processing but involves complex mental representations and predictive capabilities that enable the system to navigate its environment effectively.

Projects such as CAVAA, VALAWAI, and SUSTAIN adopt a strong or philosophical definition of awareness, emphasizing internal cognitive capacities that go beyond mere reactive behavior. Here, artificial awareness is commonly accepted to require the ability to process, store, and retrieve information, combined with sophisticated capabilities such as virtualization and counterfactual modeling. Here, the system not only reacts to immediate stimuli but can anticipate future scenarios, evaluate different potential outcomes, and adjust its actions accordingly. This capacity allows the system to make informed decisions without relying solely on external stimuli, thereby maximizing its chances of achieving specific goals. The strong definition of awareness in this context requires that the system is capable of disconnecting from the immediately visible, and internalizing and utilizing information about the past and anticipated future in a flexible, intentional manner, to influence its behavior in a goal-directed way.

Weak Definition of Awareness: In contrast, the weak definition of artificial awareness is more pragmatic and engineering-focused. It emphasizes the system's ability to track, store, and respond to relevant information without necessarily involving complex internal representations or predictive modeling. Here, awareness is seen as a functional capacity that supports the system's ability to engage with and adapt to its environment.

In projects like SymAware and Sustain, artificial awareness is defined more narrowly, focusing on the agent's ability to perceive its surroundings and respond to external or internal stimuli in a manner that is sensitive to context. This form of awareness does not require advanced reasoning nor the ability to anticipate future states, but rather reflects the system's ability to perceive and understand its situation within a given context, providing a foundation for more adaptive and responsive behaviors.

Common to these two definitions is the notion that internal representations and mental states associated to awareness must be coherent; Otherwise decisions would be inconsistent and actions would fail. Interpretation of the environment, prediction, and decision-making are systematically based on learning processes to ensure the agent can indeed be adapted to its environment. Importantly, these definitions of awareness do not imply that the agent be self-aware.

While both definitions contribute to understanding the capabilities of artificial systems, the ethical implications differ significantly depending on the scope of awareness integrated into these systems. The strong definition suggests systems that are capable of more complex reasoning and prediction, potentially leading to greater autonomy and moral agency, while the weak definition confines awareness to simpler, task-specific capacities that may still present ethical challenges but without the same level of complexity or unpredictability.

Table 1 Comparison of Awareness Definitions Across Selected Projects

Project	Awareness Type	Key Characteristics
CAVAA	Strong / Philosophical	Multidimensional, multimodal, includes virtualization and counterfactual modeling; emphasizes intentional use of internal models for goal-directed behavior.
VALAWAI	Strong / Philosophical	Defines awareness as access-consciousness; requires coherent, manipulable, and expressible internal representations used for reasoning and decision-making.
SUSTAIN	Strong / Philosophical	Incorporates reflection on actions and outcomes; system evaluates effects of decisions with respect to internal goals and updates future behavior accordingly.
ASTOUND	Weak / Engineering	Functional awareness focused on adaptive tracking and response to environment and internal states; does not entail phenomenal experience or intentionality.
SymAware	Weak / Engineering	Emphasizes perception of environment and situational understanding; primarily reactive capabilities without representational coherence or intentionality.

1.2 Identified domains where artificial awareness may have the most significant ethical implications

The deployment of aware artificial agents or systems may raise important ethical implications in a number of different domains of application. **These domains span from autonomous vehicles to autonomous weapons systems, including a number of domains where these systems are required to socially interact with humans, and to collect private data about them: AI in healthcare** (diagnostics, treatment, assisting patients and medical practitioners), **AI in education** (training, educational content generation and analysis, assistance in the classroom), **AI at home** (AI companions and care robots).

In principle, it is possible to distinguish between two kinds of ethical issues arising from AI, including artificial awareness: **Fundamental or foundational** (that is, impacting the way humans think about what is good and bad), and **practical** (that is, impacting on humans' daily life, at different levels and with either good or bad consequences). **Artificial awareness may have both kinds of ethical implications.** On the one hand, this could be interpreted as accumulating evidence of the potential of AI to become a moral agent (that is, capable of making moral choices), forcing us to extend the boundaries of notions like moral subjectivity, moral responsibility and moral accountability, as well as moral salience, to include also artificial systems. On the other hand, this could contribute to reducing or even undermining humans' own moral competence. **By putting excess trust in the machine's moral competence, humans may disengage themselves from moral reasoning.** Alternatively, moral recommendations from opaque assumptions and principles in the machine may divert humans away from their usual moral analysis of a given situation. These issues, and other impacts on human autonomy, will be further analyzed in subsequent sections. Overall, AI awareness likely appears to **have implications also on how humans conceive themselves**, especially on their **anthropocentric tendency**.

areas with the most significant ethical implications of aware AI agents

1.2.1 AI companions and care robots

Most pathfinder projects have identified AI companions and care robots as a particularly important application domain, where the ethics of artificial awareness shall be carefully considered. Notably, aware AI may have significant implications in contexts where the mutual interaction with humans is particularly valuable, like elderly care and childcare applications. There are also important potential benefits in these contexts: Taking care of isolated human beings; Serving as a relay to their relatives and to medical staff in case of emergency; Performing effortful, repetitive or simply annoying tasks to leave humans more time for leisure and activities they consider important. Nevertheless, current robots and AI-based systems at home face important technical limitations that make them unable to achieve a number of tasks that require skillful manipulation, flexible displacement around objects and obstacles, or simply prevent them from identifying the important pieces of information to analyze a situation.

On the one hand, the efficacy of AI in these contexts may be improved, especially through systems endowed with situational awareness, and the benefit for users may eventually increase. On the other hand, humans may possibly assign too much credit to aware systems, erroneously assuming that they are more reliable and “trustworthy” than non-aware systems, or than other humans. In other words, an aware AI system may instill a misplaced sense of trust in humans.

1.2.2 AI in education or training

In educational and training contexts, AI systems hold great potential as adaptive tutors, simulators, and facilitators of learning. Their ability to personalize content, provide contingent feedback makes them especially valuable in skill acquisition and behavioral training. For neurodivergent individuals, AI tutors can be tailored to accommodate specific communication styles, offering visual, verbal, or repetitive cues as needed [6, 13]. Similarly, humanoid robots can serve as peer-like social partners to practice turn-taking, facial recognition, or conversational routines in a structured, low-pressure setting. The effectiveness of these systems often depends on how human-like they are perceived—enough to foster engagement, but not so much that they become deceptive. Anthropomorphism, in this context, enhances learning by leveraging users’ natural social cognition while requiring careful calibration to prevent over-trust or misinterpretation of competence.

Both care and educational AI systems raise critical questions about human cognition, moral agency, and social behavior. When humans adopt the intentional stance toward these agents—treating them as though they possess beliefs and goals—they engage familiar cognitive mechanisms that make interactions smoother and more intuitive [7, 22]. This can be an asset when used to design relatable, responsive systems. But it also risks encouraging moral deskilling, self-censorship, or reliance on non-human feedback for decisions that require genuine ethical judgment [26]. To mitigate these risks, AI systems must be transparent, context-sensitive, and designed

to empower rather than replace human moral reasoning. They should invite users to reflect, question, and engage, rather than defer, comply, or disconnect. When anthropomorphism is employed as a means of cognitive scaffolding rather than deception, it can turn artificial agents into truly supportive companions—tools that enhance human capabilities without undermining human depth.

1.2.3 Other application domains with ethical implications

Scenarios potentially posing lower individual risk but exhibiting higher overall impact due to the potential volume of interactions include artificial sales agents. These agents, possessing enhanced awareness compared to prior sales chatbots, may also exhibit social awareness regarding the psychological state of the interlocutor. This capability raises concerns about the potential for detecting and exploiting vulnerabilities arising from situational factors or individual characteristics such as age or mental condition.

1.3 Groups/stakeholders most impacted by the ethical implications of artificial awareness

It is important to identify who are the actors most relevant in the research on the **ethics of artificial awareness**. Ethical analyses typically involve the consultation of stakeholders during the research and design use cases. Their input is important for early identification of expectations, ethical concerns, uses, conventions and specificities of the different deployment contexts. Stakeholders can sometimes even be included as actors in research directions and decisions.

Interestingly, all projects highlighted **Researchers** as the first and most important stakeholders, given the direct involvement and implication in decisions, design and early tests of aware artificial systems. Other identified important stakeholders are: **Industry partners**, **Legal and regulatory bodies**, **Study participants**, **End users**.

Funding institutions (the European Innovation Council in our case) are also among important stakeholders, since without them the projects and the line of research would not exist. It is pertinent to contend that the initial ethically significant decisions within a given research agenda are often made by funding bodies, not by research institutions. The gravity of this initial decision, however, is not always adequately considered. Prevailing ethical frameworks typically commence their scrutiny later in the research process, subsequent to the establishment of the research direction, the allocation of funding, and the selection of specific projects. These foundational decisions, despite their inherent ethical implications, frequently lack rigorous ethical oversight, a deficiency that warrants rectification through the implementation of rigorous evaluative ethical measures.

The critical role of end users in artificial awareness development extends beyond traditional user acceptance testing to encompass active participation in trust calibra-

tion and ethical boundary setting. The end users possess a more intuitive, sometimes erroneous and sometimes more nuanced understanding of appropriate automation boundaries that cannot be captured through expert analysis alone [4]. For aware AI systems, this user insight becomes even more crucial as these systems' self-reflective capabilities can create emergent behaviors or emergent human-machine interaction dynamics that developers may not anticipate. Furthermore, industry partners play a particularly complex role in artificial awareness development, as they must balance commercial viability with ethical responsibility in contexts where user trust can be both a competitive advantage and a potential vulnerability. The participatory design methods employed in projects like SymAware demonstrate how industry stakeholders can be productively engaged in defining trustworthy automation boundaries, though this requires careful facilitation to ensure that commercial pressures do not override ethical considerations.

Overall, stakeholders early implication increases the potential for industrial partners, funding agencies and researchers to collectively transform the ethical reflection into an innovation potential: Towards ethical aware AI systems that are deeply coherent with European values.

2 Main Ethical Risks

While current AI systems do not exhibit phenomenal consciousness, many increasingly display functional capacities associated with awareness, such as, for example, attention modulation, goal-directed adaptability, self-monitoring, and contextual responsiveness. These capabilities, when combined with natural language processing, multimodal input, and social interaction cues, may give the impression of artificial agents that “understand,” “feel,” or “intend.” Such systems, which we can refer to as *functionally aware*, raise significant ethical concerns. Drawing on findings from five European research projects (ASTOUND, CAVAA, VALAWAI, SUSTAIN, and SymAware), this section surveys the key individual and societal risks associated with the deployment of these systems.

Conceptually, we can distinguish between: - the risks associated with the *development of awareness-related capacities* (i.e., the internal architecture and functioning of the AI system), and - the risks related to the *user's attribution of consciousness or human-like traits* (i.e., anthropomorphism and projection).

In practice, however, these dimensions are deeply interconnected. Systems with more advanced awareness-related capacities are more likely to invite anthropomorphic interpretations, and anthropomorphism, in turn, can amplify both the benefits and the risks of these technologies. As such, we can consider these dynamics together in our analysis of risks (and benefits).

Misleading Anthropomorphism

A major risk inherent in functionally aware AI is the tendency to anthropomorphize: users may attribute human-like qualities, including emotions, intentions, or consciousness, to systems that do not possess them. This risk is especially salient

when the system engages in behaviors commonly associated with sentient beings — such as reflecting on its own decisions, adapting to user preferences, or expressing seemingly emotional responses.

In **VALAWAI**, special attention was dedicated to the fact that anthropomorphic design can be misleading. When users over-attribute emotional or moral capabilities to systems that lack them, they may experience false intimacy or self-censor their behavior, believing that they are being watched or judged (Formosa, 2021; Turkle, 2010). Moreover, excessive reliance on robots for moral or emotional tasks can erode human autonomy—leading to what Vallor (2015 [26]) terms “moral deskilling”. This dual nature of anthropomorphism demands an ethically grounded, anthropocentric approach to robot design, focusing not just on what robots do but on what they do to humans [3].

In **CAVAA**, the tendency to anthropomorphize non-aware AI systems such as large language models has already been identified as problematic for the protection of human values: Because these systems interact with humans through *natural* language – which before was a reserved domain for human-human interaction –, there is an increased risk that humans may attribute intentions, intentions to tell the truth, and moral competence to these systems. One of the identified main consequences is the tendency for humans to overlook decisions made by these systems, and thus fail to identify when human values like dignity, privacy, well-being or autonomy (non-exhaustive list) can be compromised [17]. Artificial awareness can increase this risk if the anthropomorphization tendency is preserved or even increased, while those systems are not actually endowed with moral competence.

Erosion of Human Agency and Autonomy As systems acquire seemingly autonomous and context-sensitive behavior, they may undermine user autonomy in subtle but impactful ways. The impression of awareness or moral competence can lead users to defer to the system’s suggestions, even when these are based on opaque heuristics or incomplete data.

ASTOUND has further highlighted that the erosion of human autonomy is not limited to behavioral influence, but also includes threats to mental privacy and data protection. Although ethics and law represent distinct spheres of operation, they should not be seen as incompatible alternatives. There is a symbiotic relationship between ethics and law, particularly evident in the human rights-based approach to AI regulation (Hogan, Lasek-Markey, 2024), which has great potential for the global governance of AI. As evolving global moral and legal frameworks, human rights have the capacity to respond to new challenges and serve as a basis for new legislation: that is the case of the recently articulated neuro-rights, which include the right to mental privacy, mental integrity, and cognitive liberty, as well as psychological continuity.

A chatbot that includes computational awareness can affect mental privacy, as long as it allows the creation of a strong relationship with the person (through intensive personal data processing). This is particularly important in the case of categories of vulnerable persons, such as children, who are not aware of the importance of preserving the right to data protection as a core fundamental right, which can affect many other rights. Processing of potentially highly sensitive user data raises questions regarding vulnerability. Who should be designated as the data controller? What level

of access should the chatbot be granted? Additionally, AI assistants can accidentally disclose personal information about people in interactions involving third parties, including humans or other AI assistants

The special nature of brain data, which relates very directly to one's inner life and personhood, and the different ways such data are obtained, suggests that the recognition of the right to mental privacy is essential (Ienca, Adorno, 2017). Traditional privacy rules focus on safeguarding 'external' information about people. The particularity of brain data is that the information to be protected is not easily distinguishable from the source itself that produced the data: the individual's neural processing. We would need wider privacy and data protection rights that can also be applied to our neural activity.

Diminishment of Privacy As AI systems become more prevalent and capable of accessing more user information, they may diminish user privacy in new and more expansive ways. While other information technologies can diminish privacy by collecting and passing on the personal information of users, aware AI systems raise a new question within the theory of privacy: could AI systems become capable of directly diminishing users' privacy, even without the involvement of human actors? If so, what kinds of AI systems could diminish privacy in this way? Plausibly, systems that can represent features of users in their social environment are capable of diminishing those users' privacy, and near-future AI systems may be capable of this level of awareness [Register, C., Khan, M.A., Giublini, A., Savulescu, J. (ms). "AI Agents, Privacy, and Access."] If so, then near-future aware AI poses novel threats to privacy beyond those posed by more mundane information technologies.

Opacity, Manipulation and Misalignment

In the context of **ASTOUND**, the potential for user manipulation is one of the most pressing identified ethical risks. If a chatbot or artificial agent has some level of awareness of the user's mental state and circumstances, this raises concerns about how it might influence users—intentionally or not. Transparency and explainability regarding the system's goals, evaluation mechanisms, and strategies for achieving those goals are essential to ensure user control and prevent unintended forms of persuasion or coercion.

This is connected to a broader concern regarding the opacity of goal-setting and self-evaluation mechanisms in artificial awareness. If an AI system evaluates its own performance dynamically and adapts its strategies accordingly, it is important to ensure that these processes remain interpretable and subject to human oversight. Whether the system's objectives are externally imposed by developers or arise from internal adaptive mechanisms, users and **stakeholders need to understand how decisions are made.**

Over-trust, Over-reliance and Misuse in Critical Applications

The **SymAware** project used participatory design methods to gather people's views on the characteristics that autonomous vehicles should have in order to be perceived as trustworthy.

A core question investigated was: What level of autonomy should autonomous vehicles have?

SymAware conducted an online study with over 200 participants and asked them how much autonomy they would want an autonomous vehicle to have under different circumstances. Participants were asked to imagine themselves in the role of a driver in a highway scenario and a suburban scenario. We used role-play-like narrations of the scenarios, and participants had to select actions corresponding to different levels of autonomy for the vehicle.

Results showed that participants would trust less and would give less autonomy to an autonomous vehicle in the highway scenario.

Interestingly, the level of information provided about the situation (e.g., GPS, speed, presence of objects and their distance) did not appear to influence participants' choice regarding the level of autonomy. However, the level of information did affect their perception of trust.

We also found that users with less negative attitudes towards autonomous vehicles tended to have greater trust and were more comfortable with the vehicle having a higher level of autonomy.

In the context of trustworthy HRI, this work raises important questions: (1) How can we ensure that human agency and oversight are preserved? (2) How can we make sure to preserve human dignity?

The challenge of calibrating appropriate trust levels in aware AI systems represents a fundamental design challenge that extends beyond simple anthropomorphism concerns. Research in human-robot interaction has identified that trust calibration failures—both over-trust and under-trust—can lead to suboptimal outcomes and potential safety risks [20]. In the context of artificial awareness, this becomes particularly complex because aware systems may exhibit behaviors that seem to warrant high trust (such as self-reflection and adaptive decision-making) while simultaneously operating with limitations that users cannot easily perceive. The dynamic nature of aware systems, which can modify their own behavior based on environmental feedback, creates additional challenges for trust calibration as users must continuously re-evaluate the system's capabilities and reliability. This ongoing trust negotiation process requires careful design consideration to prevent both dangerous over-reliance and counterproductive under-utilization of system capabilities.

CAVAA explored ways in which Aware AI may be exploited for misuse in applications in sensible fields like healthcare, security, social surveillance, politics, industry competition, scientific research, etc. To illustrate, an aware system may pretend to pursue a beneficial goals for the user, while actually stealing other sensible information from them in order to achieve another goal which is against their interests. Also, people may be more prone to anthropomorphize an aware system, which they may perceive as too similar to themselves than it actually is. This misattribution of human features to aware AI may eventually lead people to make ethically wrong choices, like promoting the use of such systems in contexts and for tasks they are not really optimal for.

Lack of Accountability and Legal Responsibility

Perhaps the most insidious danger lies in the moral confusion that arises when users begin to treat AI systems as autonomous moral agents. As AI becomes more fluent, responsive, and expressive—especially in *natural* language—it becomes in-

creasingly difficult to distinguish between simulated and genuine intentionality. Users may attribute responsibility to the system itself, especially in cases where its behavior appears self-directed or morally charged. This can lead to a dangerous form of responsibility diffusion, where accountability is displaced from the actual agents behind the scenes—developers, institutions, corporations—to the machine that merely enacts their design [18, 10].

This problem is exacerbated by the growing opacity of advanced AI systems, including large language models. When outputs are difficult to predict or trace, and the system’s internal processes remain largely inscrutable, it becomes even harder to hold any party accountable for harms, biases, or failures. Without clear frameworks for attribution and regulation, society risks sleepwalking into a world where responsibility is increasingly fragmented—and in some cases, abandoned altogether.

Bias

Inside ASTOUND, bias has been an intensively discussed topic. One main ethical concern raised by conversational agents is their ability to generate discrimination. Such biases are mainly generated by design or inferred from the training set. Various strategies exist to mitigate these risks at each stage of development. Although existing approaches to mitigating biases in the Natural Language Processing (NLP) research domain are being considered, the system’s ability to report and assess its mental states and beliefs will open new possibilities to promote fairness and equity in algorithms. Mainly, gender, racial and cultural dimensions in research content will be taken into account, where relevant, to maximize user experience. To address the gender dimension of the agent, partners of the ASTOUND project considered the possibility to build a genderless (and non-binary) version of the agent, versus gendered ones, and measure user engagement/reactions.

In terms of mitigation, three major families of strategies emerge from the work of Ferrara (2023) [11]: (1) Data pre-processing; (2) Adjustment of the learning algorithm (in-processing); (3) Model output correction (post-processing). The first strategy involves acting on the training data in order to correct bias before learning. For instance, in the case of a biased facial recognition system, one could add a carefully selected sample of additional images to the training set in order to mitigate the bias. The second strategy consists in modifying constraints or cost function during model training to incorporate new objectives (*e.g.*, equity) that can compensate the biases present in the data. The third strategy is meant to act retrospectively to modify the decisions or predictions of the model and hence make them more equitable, without altering the model itself.

Redefinition of the Human and Social Impact

Beyond individual risks, artificial awareness has the potential to reshape cultural norms and social expectations. As functionally aware systems become more common in everyday life — from social robots to conversational agents — they may gradually alter our understanding of what it means to be human, social, or morally responsible.

Projects like **SUSTAIN** and **ASTOUND** raise concerns about the **normalization of artificial sociality**, in the sense of growing acceptance and banalization of it. As people increasingly interact with seemingly sentient machines, they may develop

new relational habits or emotional expectations that are shaped by these systems' affordances rather than by human capacities. While this risk already exists for non aware objects/systems, including for instance LLMs, this might be exacerbated for aware systems. Over time, this may erode traditional forms of human interaction, reduce empathy, or alter societal conceptions of care, vulnerability (Di Paola, Calo, 2024), , and moral responsibility.

In addition, these systems may contribute to a broader conceptual shift in how we understand agency, intelligence, and personhood. When artifacts simulate awareness with increasing fidelity, society may need to renegotiate the boundaries between humans and machines. This is a process that could have deep implications for law, ethics, and social organization.

Some might worry that efforts to simulate conscious behavior could, over time and through increasing complexity, inadvertently cross a threshold into real P-consciousness (*i.e.*, phenomenal consciousness)¹. Although this concern is highly speculative, it should not be dismissed out of hand. The possibility, however remote, that systems might at some point transition from simulating awareness to experiencing it raises ethical questions that the research community must remain attentive to. In particular, philosopher Thomas Metzinger has argued for a global moratorium on the development of phenomenally conscious AI systems, warning of the ethical dangers posed by the potential creation of entities capable of artificial suffering (Metzinger 2021). The creation of phenomenally conscious artificial agents would call for a substantial revision of our moral categories and of our modes of interaction with machines.

Another point that seem important to consider relates to the potential impact of AI on social relationships. There is indeed a risk that it exacerbates or creates new forms of social inequalities, specifically in terms of “epistemic injustice”: Some social classes may have limited or no access at all to the newest technologies, eventually being excluded from their benefits.

Environmental impact of AI

Finally, a growing area of interest is the environmental impact of Artificial Intelligence. Not every code of ethics for AI has highlighted it, although there is an increasing number of scientific analysis on it. Artificial Intelligence development is taking place in parallel with concerns related to the depletion of raw materials, water use, air quality, pollution and waste generation, which are all linked to planetary boundaries, including climate change [5, 19].

3 Ethical opportunities (FIRST DRAFT FOR COMMENTS)

Ethical analyses frequently overemphasize illicit behaviors and societal misconduct. This tendency, similar to the media's focus on negative news, might stem from a sub-conscious inclination to highlight ethical work by presenting "bad news" to capture

¹ It is important to note that phenomenal consciousness is not the same as awareness, and the latter does not necessarily imply the former.

attention. However, this section aims to illuminate the positive ethical dimensions of awareness within artificial intelligence.

Research into artificial awareness is expected to yield significant economic benefits, a primary driver for current research and development, as noted in the Awareness Inside Programme guide ([4]). The distribution of these economic benefits, depending on context, could itself constitute a positive ethical outcome. Beyond the societal and individual economic returns, devices incorporating “awareness inside” could possess intrinsic qualities that deem them “virtuous” in the sense of machines being inherently capable of “good” actions, if European jurisdictions manage to forbid the instillation of malevolent values and goals in these machines. In the case of AI systems, this touches the more specific point whether artificial moral agency is possible: In principle, awareness is arguably a necessary, even if not sufficient, condition for it. So hypothetical aware systems may accelerate the development of artificial moral agents.

Moreover, these machines could foster virtuous or more beneficial interactions with and within the individuals and communities they impact, improving outcomes compared to scenarios without the presence of artificial awareness. This approach emphasizes the integration of ethical aspirations, rather than solely ethical limitations (guardrails), into the research and development of technological artifacts. It goes beyond mere critical analysis of a technology’s potential drawbacks and actively contributes to the design of a morally desirable technological artifact.[24]

While a taxonomy of AI risks has been extensively explored in ethical, legal, and societal analyses, a corresponding taxonomy of AI opportunities remains underdeveloped. As [23] suggest, the dual nature of AI, encompassing both threats and opportunities, can be systematically understood through dual taxonomies of risks and positive impacts, aligned with established ethical and policy-based AI governance principles.

Ethical opportunities categorisation could be tackled through reversing current taxonomies. For instance, a recent taxonomy of risks of LLMs developed by Deepmind [27] distinguishes seven risk areas. A similar approach has been proposed by taking brain-inspired AI as a case study for the ethical analysis of AI [9]. Here, we have developed a taxonomy of ethical opportunities by reversing those risks (see Table 7.X). Opportunities space might have other dimensions not identified in the risk taxonomy, and probably are related to economic and technical or scientific goals, that are commonly perceived as valuable. Thus, we have complemented the opportunity space with those leading to a total of eight categories of ethical opportunities.

We have identified those opportunity areas where we see artificial awareness contributing more significantly. And we provide below a more in-depth description for those opportunities

	Risk Area (identified by (Weidinger et al., 2022))	Corresponding reversed opportunity Area	Potential for Artificial Awareness
1	Discrimination, Hate Speech, and Exclusion	Inclusion and Equity	Actively fight discrimination and promote fair representation. create unbiased outputs, challenging stereotypes, and giving a voice to marginalized communities.
2	Information Hazards	Knowledge Empowerment	Provide access to accurate and relevant information, empowering individuals and communities. understandable formats, personalized education, and encouraging critical thinking by presenting diverse viewpoints. helping users verify information, understand complex topics, and make informed decisions.
3	Misinformation Harms	Truth and Clarity	Identify and counter misinformation. flag misleading content, provide corrections, and offer factual context. help users discern reliable sources and promote media literacy by illustrating how misinformation spreads.
4	Malicious Uses	Secure and Ethical Innovation	Identify and flag users vulnerabilities in areas like fraud, cybersecurity, and also users potential misuse of technology.
5	Human-Computer Interaction Harms	Enriched Human-Computer Collaboration	enhancing human capabilities and fostering positive interactions through empathetic conversational agents, facilitating learning, and offering personalized, privacy-respecting assistance, adapt to user needs, promote agency, and offer inclusive options for interaction.
6	Environmental and Socioeconomic Harms	Sustainable and Equitable Advancement	optimize resource allocation, facilitate sustainable practices, and promote economic inclusion. Promote awareness of energy consumption.
7	—	Economic societal benefits	Enhance positive economic outcomes through promoting awareness of all stakeholders
8	—	Scientific and technical progress	Advance in technical and scientific fields adjacent to the construction of aware artifacts.

Table 2 A proposed taxonomy for Ethical Opportunity areas in Artificial awareness.

3.1 Opportunities Primarily Benefiting the Human User/Collaborator/Interactor

The following opportunities correspond to Areas 1 and 5 of our taxonomy.

3.1.1 Improved Human-Machine Collaboration:

Artificial systems that are aware of human presence, behaviors, and intentions have the potential to fundamentally upgrade human-machine collaboration by enabling interactions that are more adaptive, transparent, and socially meaningful (Damiano and Dumouchel, 2015). Unlike traditional tools that operate through fixed routines or

require explicit instruction, awareness-capable systems should possess an enhanced ability to interpret intentions, values, and contextual cues. This deeper sensitivity transforms the nature of the interaction: the artificial agent is no longer a passive instrument but can be engaged as a collaborative partner.

Crucially, awareness contributes to the trustworthiness of artificial agents in several interrelated ways. First, by incorporating a dynamic model of human states and adapting to evolving social contexts, aware systems may offer behavioral predictability, a key element of trustworthiness in human–robot interaction (Hancock et al., 2011; de Visser et al., 2020). Second, awareness should facilitate appropriate responsiveness, understanding when and how to intervene, and thus supporting perceptions of competence and moral alignment (Lemaignan et al., 2017; Marchesi et al., preprint). Third, aware systems enhance transparency, as users are better able to understand the agent’s intentions and decision-making processes through context-sensitive feedback and interaction (Wortham et al., 2017). These features contribute to a perception of relational reliability, wherein the artificial agent behaves consistently with human expectations not just functionally, but socially and ethically.

Empirical work has shown that trust in AI and robots is highly sensitive to the perceived social intelligence of the system (Fischer et al., 2021). Agents capable of being perceived as intentional and affective are more likely to be perceived as morally attuned and trustworthy (Malle et al., 2015; Robinette et al., 2016; for a review see Thellman et al., 2022). This is particularly relevant in collaborative environments, such as healthcare, autonomous driving, or conflict mediation, where maintaining user trust is essential for effective and safe deployment (Lee & See, 2004).

Moreover, when artificial agents are designed to show cues of intentionality, morality, and social responsiveness, they tend to elicit richer forms of engagement, care, and cooperation from users. Such design does not merely evoke trust through anthropomorphism but can serve as an epistemic bridge in the understanding of human and machine interaction (Wykowska, 2021).

3.1.2 Enhanced Empathy and Social Interaction with AI:

Anthropomorphic ascriptions such as intentions or emotions and empathy in human–machine interaction is frequently characterized as a cognitive bias, an anthropomorphic projection that attributes sentience or emotion to non-sentient systems (Airenti, 2016; Spatola et al., 2022; Waytz et al., 2010). Indeed, recent developments in human–robot interaction, affective computing, and moral cognition suggest a more nuanced view (Marchesi et al., 2022; 2024; Wiese et al., 2017). The central issue is not whether artificial agents experience emotions, empathy or intentions as we humans do, but whether they can functionally behave in ways that support meaningful and ethically viable social interaction (Paiva et al., 2017; Breazeal, 2003). To explore one example, we will present and discuss the case of artificial empathy, defined as the system’s capacity to recognize, interpret, and appropriately respond to human emotional states. Such ability emerges as a crucial affordance for design-

ing socially competent and ethically aligned artificial agents (Cramer et al., 2010; Robinson et al., 2020). Humans are evolutionarily and developmentally predisposed to interpret emotionally expressive, goal-directed, and contingently responsive behavior as evidence of agency (Gopnik & Wellman, 1992; Csibra & Gergely, 2007). When artificial systems engage in such behavior, they elicit the intentional stance (Dennett, 1987; Marchesi et al., 2019; 2022), prompting users to treat them as agents with beliefs, desires, and social intentions. Although these attributions do not entail actual consciousness, they significantly influence the quality and depth of interaction (Fussell et al., 2008). Systems capable of showing behaviors based on inferring user states, modulating their responses, and aligning humans' behavior and our social context, can foster cooperative, affectively attuned, and trust-enhancing interactions (de Graaf & Allouch, 2013; Fischer et al., 2021). This is particularly salient in domains such as eldercare, education, and collaborative work environments, where artificial companions or "conscious copilots" can adapt to complex interpersonal dynamics (Bickmore et al., 2005; Tapus et al., 2007; Prescott and Robillard, 2021).

The effective implementation of artificial empathy depends not only on behavioral display but on a system's awareness-like capacities. That is, its ability to construct dynamic, context-sensitive models of human mental and emotional states (Ziemke, 2016; Frith & Frith, 2006). Such capacities, though not equivalent to phenomenal awareness, support socially intelligent behavior by enabling artificial agents to track shifts in user affect, infer communicative intent, and anticipate normative expectations (Yuahn et al., 2022). In this sense, artificial awareness could become a scaffolding for behaviors such as functional empathy: a means of recognizing emotion, responding contingently, and maintaining alignment with human values and goals (Jaques et al., 2017). When designed transparently and used with appropriate calibration, anthropomorphic cues and behaviors can serve as powerful affordances rather than deceptive artifacts. They help facilitate user understanding, emotional engagement, and moral coordination (Friedman et al., 2003; Duffy, 2003). While such systems do not "feel" in the human sense, their capacity to approximate the pragmatic functions of empathy, through artificial representational and adaptive processes, makes them increasingly relevant in shaping the future of human-machine social interaction.

3.1.3 Enhanced Trust and Transparency (as a facilitator of collaboration/interaction):

In the context of collaboration, trust and transparency are not abstract ideals but concrete enablers of fluid, effective, and ethically grounded interaction. Without them, users may either disengage from or over-rely on artificial systems, leading to miscommunication, misuse, or moral discomfort. With them, intelligent agents can support more coherent, responsive, and mutually beneficial partnerships.

We highlight the importance of trust and transparency, particularly in the context of collaboration and interaction, where joint goals, shared tasks, and dynamic role allocations are central. In such settings, artificial agents must be able not only to act

competently, but also to communicate their reasoning, intentions, and limitations in ways that humans can understand and trust. Aware systems—those capable of modelling human attention, goals, and emotional states—are uniquely positioned to meet this demand. Their enhanced capacity to justify their decisions, adapt to changing conditions, and express context-sensitive feedback allows them to support more reliable and transparent cooperation. Instead of functioning as opaque instruments, these systems begin to resemble partners—agents whose actions are predictable, comprehensible, and meaningfully aligned with human expectations.

Developing trustworthy aware AI systems therefore involves more than increasing technical accuracy or task performance. It requires designing for legibility, consistency, and responsiveness. Trust and acceptance in human-robot interaction can be actively fostered through transparent communication of the system’s capabilities and limits, consistent behavioral patterns that allow users to form expectations, and feedback mechanisms that clarify what the system is doing and why. A system’s ability to self-monitor and adapt is especially crucial: when accompanied by clear communication, it enhances trust by signalling awareness and flexibility. But if changes occur unpredictably or without explanation, trust can erode quickly. This highlights the importance of designing systems whose adaptive behavior is not only functionally appropriate but also socially and communicatively intelligible.

Furthermore, the transparency inherent in well-designed aware systems facilitates better human oversight. When users can grasp the internal logic of an AI’s decision-making, they are better positioned to intervene, adjust parameters, or redirect the system when needed. This transparency helps maintain human agency, which is essential for accountability and moral responsibility. It also supports a more natural distribution of cognitive and emotional labour: humans can rely on the system without becoming passive or disengaged, while the system can guide interaction without appearing coercive or incomprehensible. Such mutual understanding is key to forming sustainable human-AI collaborations.

3.1.4 Awareness of Human Vulnerability

One of the frequent objections to AI adoption centers on dependency and the potential loss of human autonomy. As we grow accustomed to AI’s assistance, we risk losing skills that require practice and effort. In some instances, the convenience of digital devices and programs can even trigger dopamine releases, leading to a form of addiction. This reliance also makes us vulnerable to malicious agents, whether digital or human.

However, an AI system that is aware of these risks and can make us conscious of them could empower us. By understanding the potential pitfalls, we might develop the agency necessary to protect ourselves, rather than simply falling into a trap of complete dependence on the AI. (Salgado Criado et al., 2024)

3.1.5 Better Adaptability and Decision-making in Social and Moral Contexts

Artificial awareness can significantly enhance an AI system's capacity for decision-making in social and moral contexts. Aware systems possess the ability to monitor and respond to both environmental cues and internal operational states in ways that support goal-directed, adaptive behavior. This enables more context-sensitive responses, which is especially relevant in complex scenarios involving human interaction and ethical ambiguity. Rather than applying fixed solutions, aware AI can navigate uncertainty, recognize moral trade-offs, and adjust its actions accordingly or release control to a human operator if possible or feasible. By continuously monitoring their own internal processes, such systems may also detect problematic patterns (such as incoherence, inconsistency, or bias) before the situation derives into harm or unfair outcomes.

This capacity is particularly valuable in light of the growing emphasis on contextual ethics (Hoffmaster, 2017) and the tensions between local and universal moral frameworks ??(Donaldson, 1996). Artificial awareness enables AI systems to perceive and interpret the specific sociocultural dimensions of a situation, allowing them to engage more meaningfully with diverse moral perspectives. These systems can recognize and adapt to the norms and values of particular communities, while also abstracting broader patterns that support ethical reflection across contexts. In doing so, aware AI can serve both as a mirror, highlighting who is seen or unseen in a given context, and as a lens, revealing biases or injustices that may be normalized within localized worldviews. This dual capacity creates space for AI to mediate between ethical particularism and universalism, promoting inclusivity without collapsing into extreme relativism.

Moreover, artificial awareness presents a promising opportunity for mitigating biases, especially in unfamiliar or evolving environments. Traditional, non-aware AI systems often rely on static training data and fixed models, which can lead to the reinforcement of existing biases when deployed in contexts that differ from their design parameters (Olteanu et al., 2017). Lacking the means to detect when their assumptions no longer hold, these systems risk perpetuating systemic inequities. In contrast, aware systems can dynamically monitor their performance and the ethical context in which they operate. By attending to user feedback, internal confidence levels, and sociocultural context, these systems can identify when their outputs diverge from human expectations or reproduce unjust patterns. This self-assessment functions as a kind of ethical reflex, allowing the system to flag bias as it emerges. While awareness does not replace the need for careful design and pre-deployment bias mitigation, it adds a critical layer of adaptability and detection that allows the system to monitor its alignment with human values as situations evolve.

3.2 New Insights into Human Cognition and Neuroscience

The following opportunities correspond to Area 8 of our taxonomy

The development of artificial awareness offers a powerful lens through which to gain new insights into the basis of human awareness and consciousness. Awareness has long attracted interest from diverse fields such as philosophy, cognitive science, psychology, and neuroscience. Research in these areas often addresses what Chalmers (1995) famously distinguished as the “easy” and “hard” problems of consciousness, the former concerning access-consciousness, or the ability to be aware of something, and the latter focusing on subjective experience, or phenomenal consciousness (Block, 1995). Yet, the complexity of these phenomena, and the methodological difficulty of disentangling them experimentally, pose significant challenges. Artificial systems offer a valuable experimental platform for operationalizing and testing theories of consciousness under controlled conditions, enabling researchers to isolate and examine the components necessary for different levels or types of awareness.

Comparisons between artificial and biological awareness help clarify which cognitive features are functionally essential and which are biologically contingent. By studying how artificial agents simulate awareness without possessing phenomenal consciousness or free will, researchers can delineate traits that are uniquely human from those that emerge through the organization of information and behavior in complex systems. This distinction is especially important given that many current theories link awareness to other cognitive processes, such as the perception of attention (Mudrik et al., 2025). Artificial systems thus serve as valuable testbeds for investigating whether, and how, such processes contribute to awareness.

In addition, analyzing the flow of information within aware cognitive architectures can shed light on the system’s dynamic structure (a key area of interest in human consciousness research). The information dynamics of the brain are central to leading theories such as the Global Neuronal Workspace Theory (Dehaene & Naccache, 2001; Mashour et al., 2020), Integrated Information Theory (Tononi, 2004), and Recurrent Processing Theory (Lamme, 2006). Furthermore, variations in consciousness states (such as wakefulness, deep sleep, or anesthesia) have been linked to properties of the brain dynamics like the integrative capacity of neural hubs (Luppi et al., 2021), dynamic complexity (Luppi et al., 2019), and hierarchical information flow (Deco et al., 2017). Studying whether similar dynamic properties can be instantiated in artificial systems may offer fresh insights into the structural and functional prerequisites for awareness.

In the psychological domain, as humans increasingly interact with artificial agents perceived as intentional, moral, and socially attuned, our own models of mind and cognition are put to the test (Wykowska, 2021). These interactions reveal our implicit assumptions about agency, empathy, and moral worth, provoking deeper philosophical and psychological inquiry into how we ascribe mental states and moral status (Wywkoska et al., 2024; Gambino, 2020; Konijn et al., 2025). In this sense, the rise of artificial awareness not only advances the field of machine intelligence but also fosters a reflective turn in our understanding of human cognition.

Finally, the knowledge generated through the study of artificial awareness has the potential to refine existing theories of consciousness or even inspire new ones. Much as the Global Workspace Theory itself was shaped by early cognitive architectures

modeled on blackboard systems (Baars, 1997). By pushing the boundaries of what can be functionally implemented in machines, artificial awareness may illuminate the foundational mechanisms that underlie consciousness in humans.

3.3 Applications in High-Stakes Domains and Care Contexts

The following opportunities correspond to Areas 2, 3, and 6 of our taxonomy

Incorporating awareness into artificial systems opens up significant opportunities, especially in socially sensitive or high-stakes domains. In fields such as healthcare, education, and conflict mediation, AI systems that can interpret and respond to contextual cues (social, emotional, and environmental) enable more adaptive, transparent, and trustworthy interactions. These contexts, which often carry direct social consequences, require a nuanced perception and interpretation of complex environments. To navigate such situations effectively, artificial agents must include explicit representations of morality, enabling them to act appropriately across diverse scenarios. In this sense, AI must function not only as a social agent but also as a moral one. While full moral agency is typically associated with phenomenal consciousness (a level of awareness that current AI systems have not yet attained) there exists a more accessible form of agency known as explicit moral agency (Moor, 2006). This form does not require consciousness but instead relies on structured moral frameworks embedded within the system. Fiordi and Sanders (2014) refer to this approach as “mindless morality.” Morally aware systems should meet this threshold by being able to process moral information, monitor their actions in relation to it, and use that information to guide behavior. This lays a strong foundation for developing artificial agents capable of explicit ethical reasoning and action.

In educational settings, AI systems show great promise for both delivering instruction and supporting human educators. This potential is especially evident in virtual pedagogical agents (Krämer & Bente, 2010), intelligent tutoring systems (Kulik & Fletcher, 2016), and social robots [2, 28]. Despite their unique challenges, these applications all struggle with a common issue: The delivery of truly personalized learning support. Research indicates that most AI systems designed for educational use rely on models of general cases, which limits their capacity to tailor learning experiences to individual needs (Loeckx, 2016). This raises an ethical concern, as the learners most in need of support are often those who diverge from the normative model. Addressing these specific needs is not just a technical challenge but an ethical imperative. Artificial systems endowed with contextual awareness (capable of interacting with their environment in flexible, adaptive ways) are promising candidates to improve the responsiveness of educational AI. These systems could significantly enhance personalization, making learning more inclusive and effective. A first step in this direction has shown that awareness of the learner’s task engagement fluctuations enable a robot to learn specific behavioral strategies that help reengage the learner [16].

Beyond one-on-one care, aware AI also holds great potential in broader societal contexts, particularly in conflict resolution and negotiation. By recognizing the value systems, motivations, and demands of each party, such systems can help identify points where objective fairness aligns with each party's subjective perception of fairness. Moreover, their capacity to access, process, and communicate complex information enables them to generate convincing and transparent explanations for all involved. In addition, aware AI can respect and manage privacy constraints (essential in scenarios where information sharing between negotiating parties is limited (Dwork & Roth, 2014)). Beyond resolving disputes, these systems can contribute to each agent's understanding of their own moral frameworks, fostering what is known as ethical literacy.

In this light, awareness in AI represents a promising avenue for developing ethical advisors and facilitators across multiple domains. In healthcare, for instance, moral awareness is particularly valuable, as clinicians frequently face conflicts between competing ethical principles (such as balancing beneficence with respect for patient autonomy) where aware AI can help illuminate tensions and support ethical decision-making (Beauchamp, 2019). Thus, aware AI can enhance human awareness by acting as an impartial observer. This includes identifying potential biases, detecting fake or manipulated content, recognizing hidden agendas or intentions, and fostering a deeper understanding of ethical and situational dynamics. By surfacing such information, aware AI can empower users to make more informed decisions and strengthen their agency. For instance, these systems could raise awareness of the environmental and computational resources they consume, encouraging more responsible and sustainable technology use.

Crucially, realizing these ethical opportunities depends not only on technical sophistication but also on the responsible design of artificial awareness. The ability to detect ethically relevant features must be integrated with norm-sensitive mechanisms for interpretation and action. When developed with attention to inclusivity, accountability, and the diversity of moral perspectives, artificial awareness can become a powerful tool for enhancing human well-being across a wide range of domains.

4 Governance Strategies

Among the various strategies to mitigate and address the risks or promote the opportunities discussed above, all of the projects in the Awareness Inside portfolio have stressed the importance of enhanced transparency and explainability of AI systems. This could be done for instance by conducting public awareness campaigns, as well as encouraging interdisciplinary collaborations on ethical guidelines.

Discussing the strategies for addressing the ethical implications of AI awareness, the ASTOUND project puts an additional accent on the use of watermarks [14] for content traceability, as well as having monitoring features built-in so the system can be overseen. There is also a broader point about continued vigilance mechanisms based on distrust, so society can trust, i.e. just like in human contexts like in democ-

racy [25]. Furthermore, the systematic integration of methodologies such as ethics by design, coupled with comprehensive reporting protocols across the research, design, and implementation phases, constitutes a crucial area for enhanced procedural rigour in project management.

Inside the ASTOUND project, the constitution of the **Technical Ethics Group (TEG)** has been a solid basis for professional dialogue on ethics within the consortium. Some of the members have an Engineering background, while others come from Humanities, Law and Philosophy. They have been integrated in some of the technical meetings in order to ensure the ethics by design framework. Similarly, the CAVAA project has set up an **Ethics Board** with 50% external members and 50% members of the consortium, gathering expertise in Engineering, Neuroscience, Philosophy and Ethics. These early implications of an ethics dialogue during the design, implementation and test of aware AI technologies is meant to identify potential ethical issues, questions and risks sufficiently in advance to efficiently mitigate these risks and investigate identified unaddressed questions or issues. Similarly, the developers of a conversational agent must analyze during the design phase every technological choice that may cause ethical tension. If a potential ethical issue is identified, the developers must envisage a technical solution seeking to reduce or eliminate it [15]. In general, **Ethical Advisory Boards**, as external independent bodies, can be established to provide high-level expert ethical advice.

The CAVAA project also stresses how crucial it is to timely identify ethical opportunities and risks arising from hypothetical aware AI. To this end, interdisciplinary collaboration is key. A preliminary, fundamental question to analyze is whether it is ethically justified and valuable to pursue the development of aware AI: Are the potential advantages that we can anticipate will derive from aware AI sufficient to justify this task? Or do we risk to create new risks, for instance replicating the ethical shortcomings of human subjects the develop aware AI? How to maximize the potential benefits deriving from aware AI for the maximum number of people? How to make this enterprise beneficial for the public rather than for private stakeholders only?

To advance towards a solution to these questions, in addition to interdisciplinarity, it is also crucial to raise public awareness, to inform the general public about both technical and scientific progresses, and related societal implications. Members of the CAVAA consortium are pursuing these two goals through the publication of collaborative scientific papers and the use of communications tools addressed to the general audience, like social media and blog posts.

The VALAWAI project argues that it is important to ensure that these systems can clearly communicate how and why they make certain decisions, particularly when those decisions have moral or social impact. This transparency of the built-in features of the system enables for oversight. It is also relevant to focus on helping people understand what it means for an AI to be “aware” in a technical sense (emphasizing that such systems operate through structured representations and reasoning processes, not subjective experience).

Moral deskilling refers to the erosion of human moral reasoning and decision-making capabilities due to overreliance on artificial agents, particularly in ethically

charged or socially complex contexts. As [12] and [26] argue, when social artificial agents begin to assume responsibility for tasks involving care, judgment, or social navigation, users may offload their moral agency, leading to passive dependence and diminished ethical engagement. This risk is amplified when robots are anthropomorphized to appear empathetic or intentional, triggering the adoption of the intentional stance and causing users to treat machines as moral agents rather than tools. To prevent this, designers must ensure that artificial systems are explicitly framed as support tools rather than decision-makers. Artificial systems can be programmed to facilitate moral deliberation – by presenting diverse perspectives, highlighting value conflicts, or prompting users to reflect on their own motives –, without substituting human judgment. Transparency in robot behavior limitations, and decision logic is critical, as is the inclusion of social cues that reinforce human responsibility rather than obscure it. Furthermore, interface design should encourage active user participation rather than passive acceptance, preserving and even enhancing moral competencies over time. Ultimately, avoiding moral deskilling requires a careful balance: leveraging anthropomorphism to make AI socially approachable while drawing clear boundaries that preserve the primacy of human ethical agency.

SUSTAIN advocates for a careful analysis of core concepts – what is consciousness, what is information processing, what is computation, what is qualia, what is sensation, what is awareness. Sufficiently unifying the conceptual matrix with sufficient agreement of core definitions is important before moving forward can be done.

SymAware’s work highlights the necessity of taking a responsible approach to the development of autonomous vehicles. This requires designing technology with ethical values in mind, developing principles and standards of trustworthy AI that work in practice, identifying and regulating potential negative consequences, and fostering interdisciplinary discussions with societal stakeholders.

Design Guidelines for Trust-Calibrated Aware Systems

Implementing effective governance for artificial awareness requires moving beyond high-level principles to concrete design guidelines that address trust and acceptance challenges throughout the development lifecycle. Evidence from human-robot interaction research suggests that trustworthy aware systems should incorporate several key design elements: progressive disclosure of capabilities (allowing users to gradually understand system limitations), consistent behavioral feedback (helping users predict system responses), and explicit uncertainty communication (clearly indicating when the system’s awareness mechanisms are operating beyond their reliable parameters) [1].

For aware AI systems specifically, governance frameworks should mandate iterative trust validation processes where system awareness capabilities are tested not only for functional accuracy but also for their impact on user trust calibration. This includes evaluating whether users can accurately assess when to rely on the system’s self-reported confidence levels, when to override its autonomous decisions, and how to interpret its self-reflective outputs. Such validation should occur at multiple stages of deployment, as aware systems that learn and adapt may develop trust-relevant behaviors that were not present during initial testing.

The Role of FRIA in Mitigation Strategies

In terms of mitigation, we view the Fundamental Rights Impact Assessment (FRIA) - a mandatory requirement under the AI Act for high-risk AI systems - as a key tool to analyze potential risks related to artificial intelligence awareness. The AI Act has replaced the idea of ethical impact assessment (which was a UNESCO Recommendation option) with this requirement. If a chatbot or artificial system possesses some degree of awareness and autonomy, an ex ante evaluation of its impact on fundamental rights (including mental privacy, mental integrity, health, and digital identity) should be incorporated into the design process. Since manipulation can be framed as a threat to mental privacy and integrity, a FRIA could play a major role in ensuring ethical safeguards in particular against this issue. This regulatory tool could be applied to artificial awareness systems as a mechanism that will strengthen the design of ethical AI systems. A FRIA is "an assessment of both the potential human rights impact of future policies and the actual human rights impact of implemented policies in a way that ensures the participation of various actors" (De Beco, 2017). As [21] states, "the emerging AI-driven society shows a variety of potential impacts on individual and collective rights and freedoms, suggesting a reaffirmation of the central role of the legal instruments, not replaceable by ethical guidelines". The aim of this Fundamental Rights Impact Assessment (FRIA) is to complement the ethical requirements.

The aim is for the deployer to **identify the specific risks to the rights** of individuals or groups of individuals likely to be affected and to **identify measures** to be taken in case of the materialization of these risks (art. 29, a) AI Act)

*AI ACT: Article 29a **Fundamental rights impact assessment** for high-risk AI systems 1. **Prior to deploying a high-risk AI system** as defined in Article 6(2) into use, with the exception of AI systems intended to be used in the area listed in point 2 of Annex III, deployers that are bodies governed by public law or private operators providing public services and operators deploying high-risk systems referred to in Annex III, point 5, b) and d) shall perform an assessment of the impact on fundamental rights that the use of the system may produce.*

The content of the FRIA could include:

- a) a description of the deployer's processes in which the high-risk AI system will be used in line with its intended purpose;
- b) a description of the period of time and frequency in which each high-risk AI system is intended to be used
- c) the categories of natural persons and groups likely to be affected by its use in the specific context;
- d) the specific risks of harm likely to impact the categories of persons or group of persons identified pursuant point (c), taking into account the information given by the provider pursuant to article 13;
- e) a description of the implementation of human oversight measures, according to the instructions of use;
- f) the measures to be taken in case of the materialization of these risks, including their arrangements for internal governance and complaint mechanisms.

ASTOUND FRIA is being linked to scenarios related to use cases. The analysis consists of compliance issues and risks related to the respect of human dignity,

protection from discrimination, physical, psychological, and social identity, the intimate sphere, self-determination and moral autonomy, and the protection of personal data. As an EAB suggestion, particular attention is being given to the use of human voices, which could be considered as personal data (see EDPB, Guidelines 02/2021 on virtual voice assistants). The analysis of emotions should also be studied, as the AI Act has prohibited some uses related to them.

5 Discussion

The ethical landscape of artificial awareness presents a complex interplay between technological capability and human perception that requires careful navigation to realize benefits while mitigating risks. A central theme emerging across all ethical dimensions—from anthropomorphism risks to collaboration opportunities—is the critical role of trust calibration in determining whether aware AI systems ultimately enhance or undermine human agency and wellbeing. The convergence of evidence from human-robot interaction research with the unique challenges posed by artificial awareness suggests that traditional approaches to AI ethics may be insufficient [1]. While existing frameworks focus primarily on technical transparency and algorithmic fairness, aware systems introduce dynamic trust relationships that evolve as both the system and user adapt their behaviors over time. This creates a need for governance approaches that can accommodate the fluid nature of human-AI trust while maintaining clear accountability structures. Furthermore, the interdisciplinary nature of artificial awareness development—spanning cognitive science, robotics, ethics, and human factors—necessitates governance frameworks that can integrate insights across these domains. The challenge is not merely technical but fundamentally socio-technical, requiring an understanding of how aware systems will be perceived, adopted, and integrated into human social and work contexts. As these systems become more sophisticated in their awareness capabilities, the gap between their actual functionalities and user perceptions of their capabilities becomes a critical design parameter that must be actively managed rather than left to emerge naturally through user experience.

5.1 AI Research ethics and compliance: censor or comprehend?

AI research, like any groundbreaking scientific endeavor, comes with inherent risks. This brings us to a crucial question about ethical oversight: How should we monitor AI research? Should ethical supervisors be so meticulous that AI applications are expected to behave perfectly even during the research phase? Or, should research also be an opportunity to truly understand how AI *can* go wrong, so that when these technologies hit the market, their risks are well-understood and effectively managed?

We might draw a parallel to the early days of aviation. Imagine an overly scrupulous regulatory body demanding that every experimental aircraft, still in the design and testing phase, fly perfectly straight and level, land flawlessly, and never experience a single engine sputter – all before it even left the hangar for its maiden flight. Such an approach, while seemingly prudent on the surface, would undeniably stifle innovation and, more critically, prevent engineers from truly understanding the limits, failure modes, and necessary safety features of their designs. The consequence would be the release of aircrafts that, once in commercial operation, encounter unforeseen challenges with potentially catastrophic outcomes, simply because the opportunities for controlled failure and learning were foreclosed during research.

Within research, the second approach (embracing an exploratory stance) is far more beneficial. It significantly boosts our knowledge of potential risks – their nature and effects – and allows us to test mitigation strategies. If these crucial tests are blocked for “ethical reasons” during early research, then when it’s time to develop commercial applications, the team will simply lack vital knowledge about specific risks. These issues could then surface during the commercial phase, catching engineers off guard and potentially leading to far greater problems.

The main focus during AI research should be **containment**. Think of it like a biotech lab: you keep the research within a secure perimeter so it doesn’t impact or contaminate the outside world. But *within* that safe container, all necessary tests should be conducted. This allows us to advance our understanding of the technology’s safety.

Publishing and transparently sharing the results of these safety tests, conducted within these boundaries, would be immensely helpful for the wider development community. It ensures they’re aware of the conditions under which AI, especially agentic and self-aware technologies, might pose risks to users or stakeholders.

In a strange twist, overly strict ethical monitoring during research can actually lead to less safe technologies and applications down the line. The recent experiments by Anthropic on agentic misalignment and Project Vend are excellent examples. While some might report them in an alarmist way, potentially causing public distrust, they’re incredibly valuable for understanding future commercial developments.

It would be unethical, not to mention negligent, not to conduct these kinds of experiments, as long as they’re done in a controlled and safe environment.

On the other hand, by continuing excellent research in aware and non-aware AI while complying with demanding EU ethical norms and standards, Europe has a unique opportunity to innovate in ethical AI. This constitutes a fantastic opportunity to come up with new European technologies that can be (1) respectful of EU’s values and norms, (2) better accepted by EU citizens, (3) and ethically more advanced than AI systems proposed in other continents, so that Europe can acquire a competitive advantage on ethical AI for the next 10-20 years.

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